

Group Project Assignment

Course: Text, Image and Video Analytics using AI

Evaluation Component: Group Project

Weightage: 40% of Total Grade

Timeline: Final Submission due one day before Sessions 17–18

Project Title: Multimodal AI Strategy for EV (Bus) Market Disruption

1. Executive Brief & Context

The Electric Vehicle (EV) landscape, especially buses, has now entered the conversation. Winners in this hyper-competitive market are no longer determined purely by engineering specs. In fact, they are now decided based on how effectively they extract strategic intelligence from massive streams of unstructured customer data.

In this assignment, your group will act as a **Strategic AI Taskforce** for an automotive OEM. Your objective is to process a vast, complex **multimodal dataset** (text, images, and video assets) to unlock deep consumer insights, evaluate visual branding assets, analyze digital video marketing engagement, and construct a functional **Multimodal RAG (Retrieval-Augmented Generation) System** that serves as an executive decision-making tool.

2. Dataset Architecture (The Multimodal EV Portfolio)

Your team must compile and analyze a localized dataset consisting of the following four data

1. The Text Package:

- A compiled corpus of **user reviews and social media discussions** spanning multiple platforms, capturing "pre-purchase, purchase, and post-use" cycles.

2. The Image Package:

- A repository of **images** representing user-generated content (UGC), competitor EV product designs, dashboard UI/UX screenshots, and infrastructure/charging station configurations.

3. The Video Package:

- A curated playlist of **high-impact videos**, including comprehensive YouTube influencer reviews, official OEM television commercials, and technical test-drive footage.

4. The Multimodal Reference Package:

- A compilation of unstructured textual and visual sources.

3. Phased Milestone Tasks

Phase 1: Text Analytics (Module 1)

Your team will build a text analytics pipeline to dissect what consumers value or fear about EV adoption.

- **Task 1.1 (Sentiment Classification):** Preprocess your text corpus and execute a comparative sentiment analysis. You must build a pipeline to classify sentiment orientations across the pre-purchase and post-use lifecycle.
- **Task 1.2 (Topic Modeling):** Implement topic modelling to discover latent consumer discussion themes.
- **Strategic Output:** Come up with insights to guide immediate product improvements and communication positioning.

Phase 2: Image & Video Analytics for Brand Optimization (Modules 2 & 3)

- **Task 2.1 (Image Analytics):** Extract low-level and high-level visual features from your image repository. Identify what physical features correlate with perception.
- **Task 2.2 (Video & Engagement Analytics):** Analyze your video corpus by evaluating structural and pacing metrics and map them against public engagement metrics (views, likes, shares).
- **Strategic Output:** Deliver an Ad-Campaign Optimization Playbook detailing exactly what visual aesthetics and video narrative structures drive peak digital engagement for the target EV audience.

Phase 3: Building the Multimodal RAG Pipeline & Executive Strategy (Module 4)

You will synthesize all unstructured files into a consolidated strategic engine.

- **Task 3.1 (Pipeline Engineering):** Design and execute an operational **Multimodal RAG System**. Your architecture must ingest your text assets, Statista charts, and IEA market trends, utilizing cross-modal embeddings to index both textual data and visual layouts.
- **Task 3.2 (Executive Query Testing):** Your RAG engine must successfully handle multi-layered strategic queries.
- **Strategic Output:** Relying on the outputs of your RAG pipeline, compile a **Comprehensive 3-Year Corporate EV Strategy**. This strategy must detail market positioning, infrastructure expansion partnerships, battery supply chain risk mitigations, and digital marketing spend optimization.

4. Deliverables & Evaluation Weights

The project grade will be split across three distinct artifacts to ensure both technical rigor and excellence in strategic output:

Component	Format	Evaluation Focus	Weight
Technical Notebook & Code	Google Colab / GitHub Repository	• Code cleanliness, • Text/image/video pipeline implementations, • Hyperparameter tuning, • RAG pipeline orchestration.	50%
Strategic Business Report	Executive PDF Document (Max 20 Pages)	• Translation of AI metrics into corporate strategy, • Robust data-driven visualization, • Comprehensive market analysis.	20%
Boardroom Pitch Presentation	Live Presentation + Q&A	• Defense of analytical models, • Cohesive strategic vision during the final sessions.	30%

5. Evaluation Rubric Criteria

Your submissions will be graded based on four pillars:

1. **Multimodal Technical Depth:** Demonstrated mastery in processing text, image, and video data through independent pipelines before passing them into a functional RAG environment.
2. **Analytical & Strategic Cohesion:** The ability to convert machine learning metrics (e.g., sentiment polarity coefficients, bounding box frequencies, topic weights) into real-world business strategy choices.
3. **Methodological Rigor:** Proper data preprocessing, sound validation techniques, awareness of AI biases, and robust evaluation metrics for model performance.
4. **Professional Execution:** Clear business communication, well-structured visualizations, and an executable, data-backed boardroom presentation.

Submission Stages & Timelines

- **1 July (EOD):** 1 Page Project Proposal (Problem Statement, relevance, stakeholders who will be impacted by the insights, hypotheses) and Tentative Data Sources. 1 Page References. [Email to deep.prakash@spjmr.org, CC group members and the TA of the course]

- **By 5 July (EOD):** Schedule a call with Mr Param to better understand the nuances of the identified problem statement. Different ways to explore it and make it more focused.
- **By 8 July (Session 9):** Keep the text analysis results and pipeline ready for in-person discussion when Mr Param is on campus for the guest lecture.
- **Final Submission 18th July (EOD)** [Wisenet link will be shared]